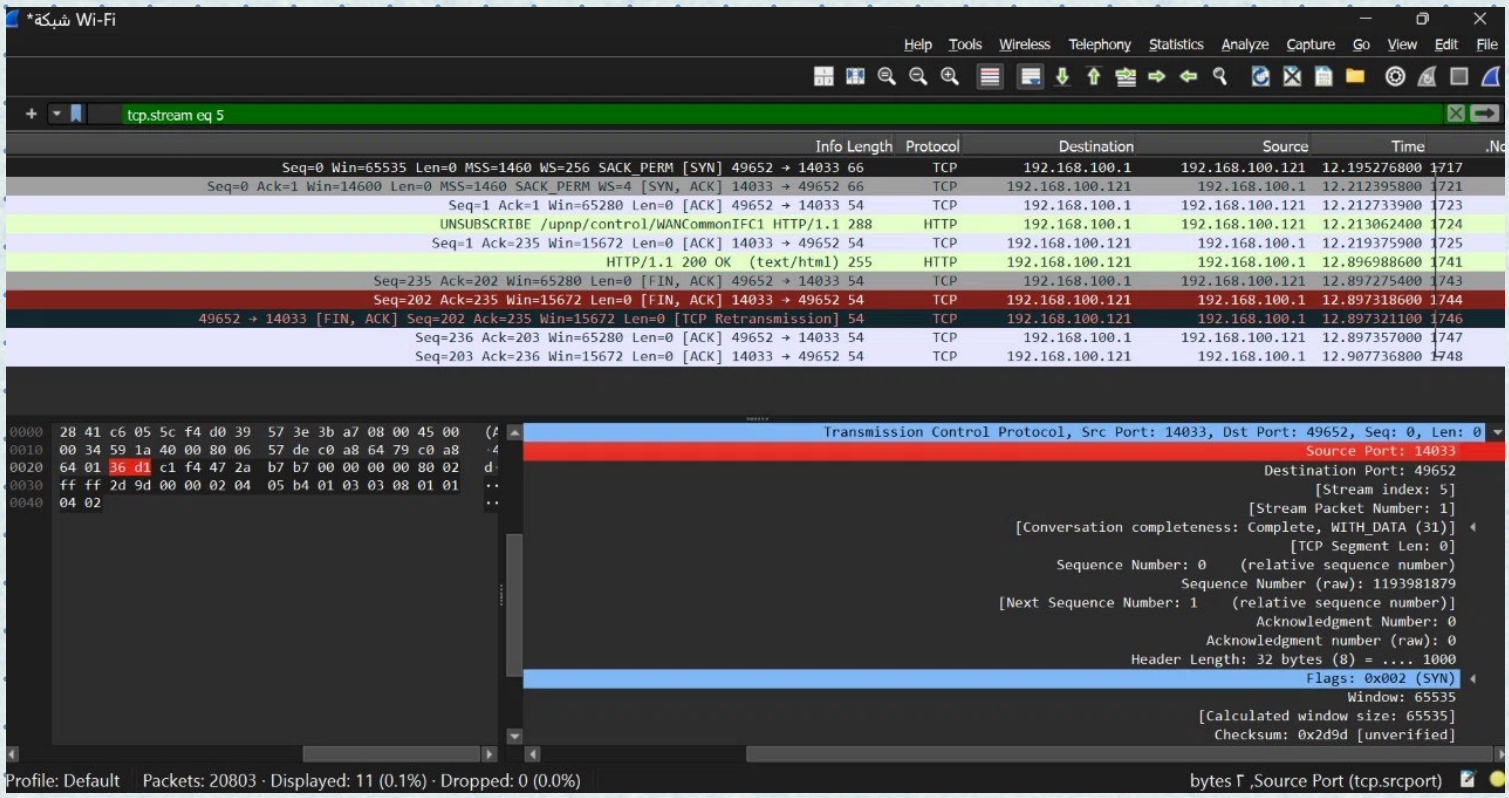
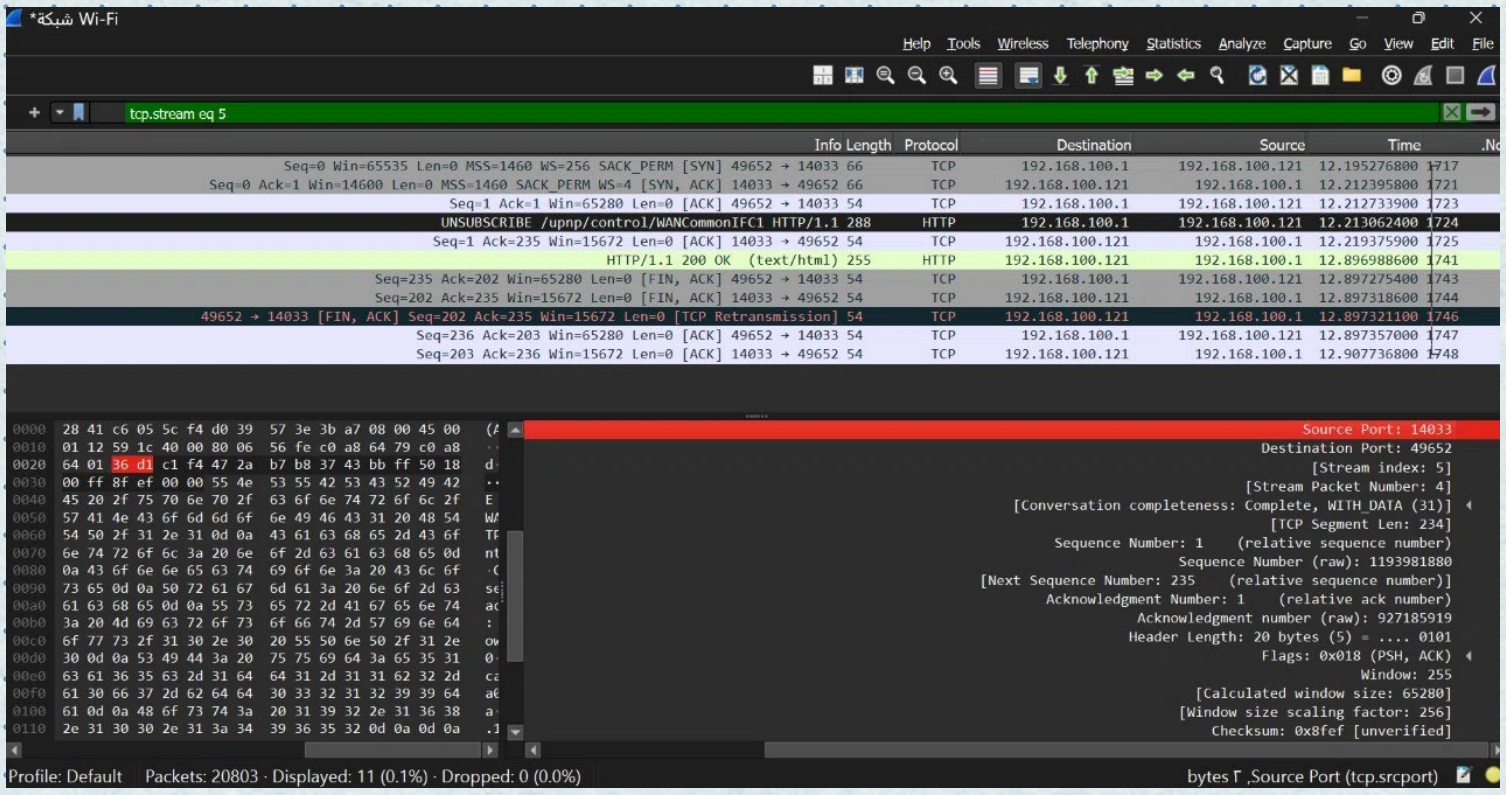

*شبكة Wi-Fi
Help Tools Wireless Telephony Statistics Analyze Capture Go View Edit File
tcp.stream eq 5
Info Length Protocol Destination Source Time
Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [SYN] 49652 → 14033 66 TCP 192.168.100.1 192.168.100.121 12.195276800
Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=4 [SYN, ACK] 14033 → 49652 66 TCP 192.168.100.121 192.168.100.1 12.212395800
Seq=1 Ack=1 Win=65280 Len=0 [ACK] 49652 → 14033 54 TCP 192.168.100.1 192.168.100.121 12.212733900
UNSUBSCRIBE /upnp/control/WANCommonIFC1 HTTP/1.1 288 HTTP 192.168.100.1 192.168.100.121 12.213062400
Seq=1 Ack=235 Win=15672 Len=0 [ACK] 14033 → 49652 54 TCP 192.168.100.121 192.168.100.1 12.219375900
HTTP/1.1 200 OK (text/html) 255 HTTP 192.168.100.121 192.168.100.1 12.896988600
Seq=235 Ack=202 Win=65280 Len=0 [FIN, ACK] 49652 → 14033 54 TCP 192.168.100.1 192.168.100.121 12.897275400
Seq=202 Ack=235 Win=15672 Len=0 [FIN, ACK] 14033 → 49652 54 TCP 192.168.100.121 192.168.100.1 12.897318600
49652 → 14033 [FIN, ACK] Seq=202 Ack=235 Win=15672 Len=0 [TCP Retransmission] 54 TCP 192.168.100.121 192.168.100.1 12.897321100
Seq=236 Ack=203 Win=65280 Len=0 [ACK] 49652 → 14033 54 TCP 192.168.100.1 192.168.100.121 12.897357000
Seq=203 Ack=236 Win=15672 Len=0 [ACK] 14033 → 49652 54 TCP 192.168.100.121 192.168.100.1 12.907736800
0000 28 41 c6 05 5c f4 d0 39 57 3e 3b a7 08 00 45 00
0010 00 34 59 1a 40 00 80 06 57 de c0 a8 64 79 c0 a8
0020 64 01 36 d1 c1 f4 47 2a b7 b7 00 00 00 00 80 02
0030 ff ff 2d 9d 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02
Transmission Control Protocol, Src Port: 14033, Dst Port: 49652, Seq: 0, Len: 0
Source Port: 14033
Destination Port: 49652
[Stream index: 5]
[Stream Packet Number: 1]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 1193981879
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
Header Length: 32 bytes (8) = ... 1000
Flags: 0x002 (SYN)
Window: 65535
[Calculated window size: 65535]
Checksum: 0x2d9d [unverified]
Profile: Default
Packets: 20803 · Displayed: 11 (0.1%) · Dropped: 0 (0.0%)
bytes Γ ,Source Port (tcp.srcport)


*شبكة Wi-Fi
Help Tools Wireless Telephony Statistics Analyze Capture Go View Edit File
tcp.stream eq 5
Info Length Protocol Destination Source Time
Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [SYN] 49652 → 14033 66 TCP 192.168.100.1 192.168.100.121 12.195276800
Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=4 [SYN, ACK] 14033 → 49652 66 TCP 192.168.100.121 192.168.100.1 12.212395800
Seq=1 Ack=1 Win=65280 Len=0 [ACK] 49652 → 14033 54 TCP 192.168.100.1 192.168.100.121 12.212733900
UNSUBSCRIBE /upnp/control/WANCommonIFC1 HTTP/1.1 288 HTTP 192.168.100.1 192.168.100.121 12.213062400
Seq=1 Ack=235 Win=15672 Len=0 [ACK] 14033 → 49652 54 TCP 192.168.100.121 192.168.100.1 12.219375900
HTTP/1.1 200 OK (text/html) 255 HTTP 192.168.100.121 192.168.100.1 12.896988600
Seq=235 Ack=202 Win=65280 Len=0 [FIN, ACK] 49652 → 14033 54 TCP 192.168.100.1 192.168.100.121 12.897275400
Seq=202 Ack=235 Win=15672 Len=0 [FIN, ACK] 14033 → 49652 54 TCP 192.168.100.121 192.168.100.1 12.897318600
49652 → 14033 [FIN, ACK] Seq=202 Ack=235 Win=15672 Len=0 [TCP Retransmission] 54 TCP 192.168.100.121 192.168.100.1 12.897321100
Seq=236 Ack=203 Win=65280 Len=0 [ACK] 49652 → 14033 54 TCP 192.168.100.1 192.168.100.121 12.897357000
Seq=203 Ack=236 Win=15672 Len=0 [ACK] 14033 → 49652 54 TCP 192.168.100.121 192.168.100.1 12.907736800
0000 28 41 c6 05 5c f4 d0 39 57 3e 3b a7 08 00 45 00
0010 01 12 59 1c 40 00 80 06 56 fe c0 a8 64 79 c0 a8
0020 64 01 36 d1 c1 f4 47 2a b7 b8 37 43 bb ff 50 18
0030 00 ff 8f ef 00 00 55 4e 53 55 42 53 43 52 49 42
0040 45 20 2f 75 70 6e 70 2f 63 6f 6e 74 72 6f 6c 2f
0050 57 41 4e 43 6f 6d 6d 6f 6e 49 46 43 31 20 48 54
0060 54 50 2f 31 2e 31 0d 0a 43 61 63 68 65 2d 43 6f
0070 6e 74 72 6f 6c 3a 20 6e 6f 2d 63 61 63 68 65 0d
0080 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 43 6c 6f
0090 73 65 0d 0a 50 72 61 67 6d 61 3a 20 6e 6f 2d 63
00a0 61 63 68 65 0d 0a 55 73 65 72 2d 41 67 65 6e 74
00b0 3a 20 4d 69 63 72 6f 73 6f 66 74 2d 57 69 6e 64
00c0 6f 77 73 2f 31 30 2e 30 20 55 50 6e 50 2f 31 2e
00d0 30 0d 0a 53 49 44 3a 20 75 75 69 64 3a 65 35 31
00e0 63 61 36 35 63 2d 31 64 64 31 2d 31 31 62 32 2d
00f0 61 30 66 37 2d 62 64 64 30 33 32 31 32 39 39 64
0100 61 0d 0a 48 6f 73 74 3a 20 31 39 32 2e 31 36 38
0110 2e 31 30 30 2e 31 3a 34 39 36 35 32 0d 0a 0d 0a
Source Port: 14033
Destination Port: 49652
[Stream index: 5]
[Stream Packet Number: 4]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 234]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1193981880
[Next Sequence Number: 235 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 927185919
Header Length: 20 bytes (5) = ... 0101
Flags: 0x018 (PSH, ACK)
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0x8fef [unverified]
Profile: Default
Packets: 20803 · Displayed: 11 (0.1%) · Dropped: 0 (0.0%)
bytes Γ ,Source Port (tcp.srcport)

Step 5: Compare the simplicity of UDP headers with TCP headers.

Just note that UDP has minimal, simple details—ports, length, and checksum—compared to TCP, where we saw sequence numbers, acknowledgment numbers, flags, and so on.

Part 4: Comparing TCP and UDP by filling in the following tables. Save your work (e.g., in an MS Word document), and upload it to your online git repo.

Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	**TCP**	**TCP is reliable and connection-oriented. It establishes a connection using a three-way handshake before data transmission.**
Data Integrity and Ordering	**TCP**	**TCP ensures data integrity and delivers packets in the correct order.**

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	**Web browsing, email, and file transfer.**	**Video streaming, VoIP, and online gaming.**
Performance	**Slower due to connection establishment .acknowledgments, and error checking**	**Faster with lower overhead because it does not require connection establishment or acknowledgments.**